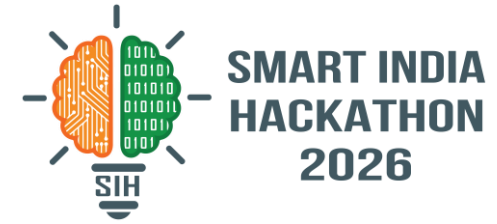


SMART INDIA HACKATHON 2026 KAVACH



Sovereign Agentic AI Workbench for Confidential Industrial Work

Problem Statement ID: **SIH26117**

Theme: Smart Automation

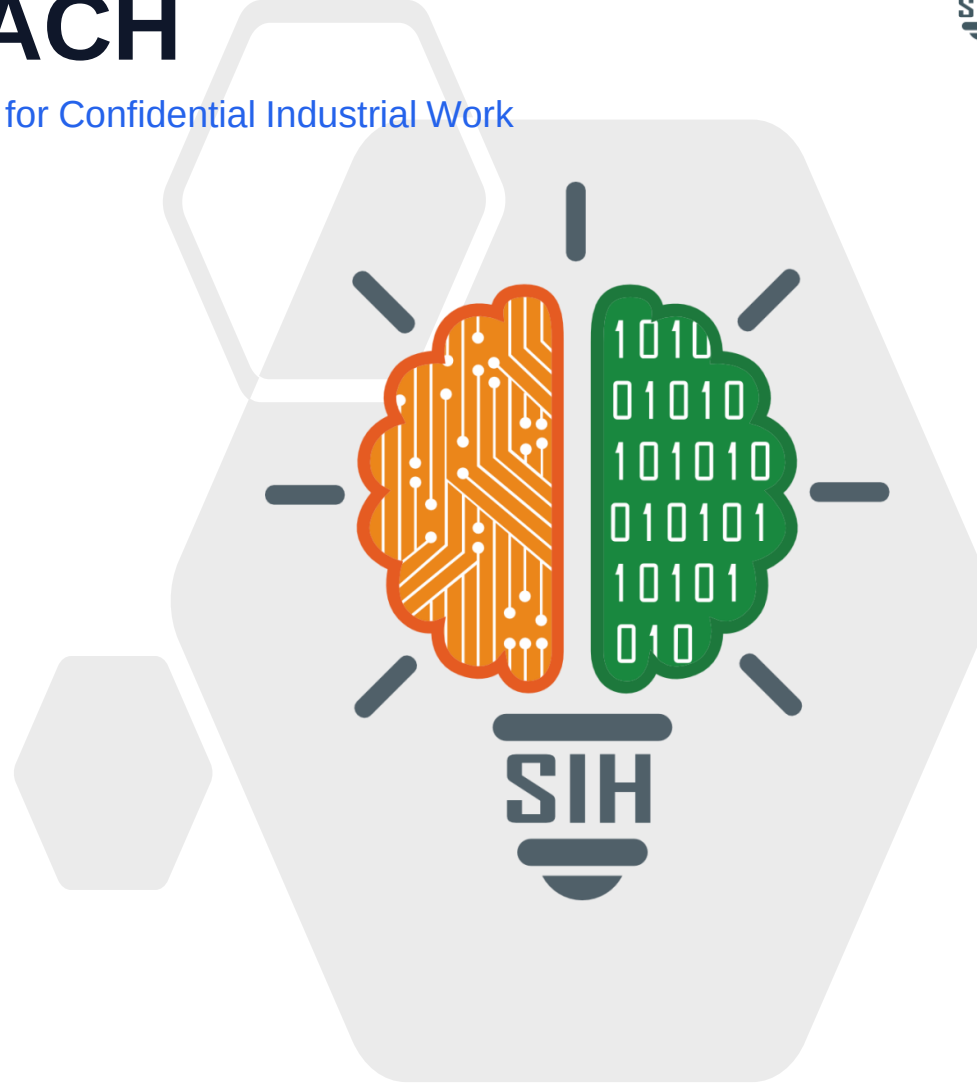
PS Category: Software

Organization: **Mangalore Refinery & Petrochemicals Ltd. (MRPL)**

Team ID: [TEAM ID]

Team Name: [TEAM NAME]

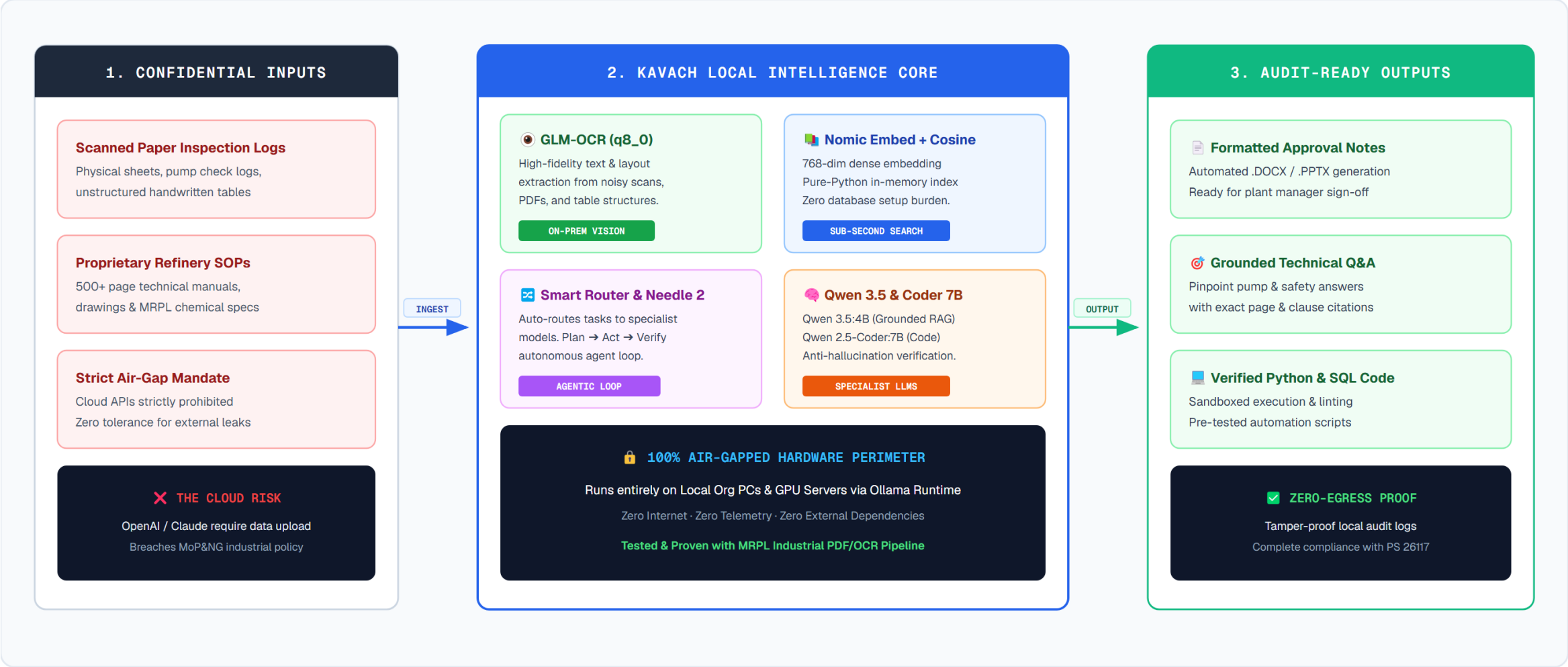
"Claude/Codex-grade intelligence. Zero data leaves the fence."



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Name

PROPOSED SOLUTION: KAVACH SOVEREIGN WORKBENCH

From Confidential Industrial Data to Verified Deliverable — 100% On-Premise



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TECHNICAL APPROACH: AIR-GAPPED SYSTEM ARCHITECTURE

End-to-End Multimodal Layer Stack with Zero External Data Egress



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AIR-GAP SECURITY GUARANTEE

- ✗ NO Cloud APIs (No OpenAI/Azure)
- ✗ NO External Outbound Telemetry
- ✓ 100% Sovereign Offline Operation

HARDWARE & BENCHMARKS

Minimum Workstation HW:
Intel i7 / Ryzen 7 · 16GB RAM · No GPU required

Production Server Specs:
1x NVIDIA RTX 3060/4090 or A100 (Sub-sec inference)

Retrieval Latency:
< 150ms Cosine similarity search over 1000+ chunks

Economic ROI:
₹0 per-token billing · Millions saved annually

Target PS: SIH26117 (MRPL Refinery)

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FEASIBILITY, RISK MITIGATION & SCALABILITY ROADMAP

Practical On-Premise Deployment Validated for Industrial Infrastructure



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⚙️ Technical Feasibility

- Open-weight models deployable locally today via Ollama runtime.
- Runs efficiently on standard workstations (Intel i7/i9, 16-32GB RAM).
- In-memory cosine vector store eliminates database setup burden.

🚀 Deployment Feasibility

- Standalone MVP operable on single industrial PC or GPU workstation.
- Seamlessly scales to multi-GPU server clusters without code change.
- 100% offline installation; air-gapped LAN deployment ready.

🛡️ Operational Feasibility & Risk Strategy

- Risk: CPU latency → Mitigation: 4-bit/8-bit quantization (Q8_0/Q4_K_M).
- Risk: Document noise → Mitigation: GLM-OCR structured text filtering.
- Risk: Hallucination → Mitigation: Strict grounded prompt with source citations.

✅ Scalability & Future Growth

- Modular model swapping: upgrade to 14B/32B models as hardware scales.
- Extendable tool catalogue: add Python sandbox, SQL connectors, CAD viewers.
- Enterprise scale: Single PC → On-Prem Server → Multi-User Refinery Cluster.

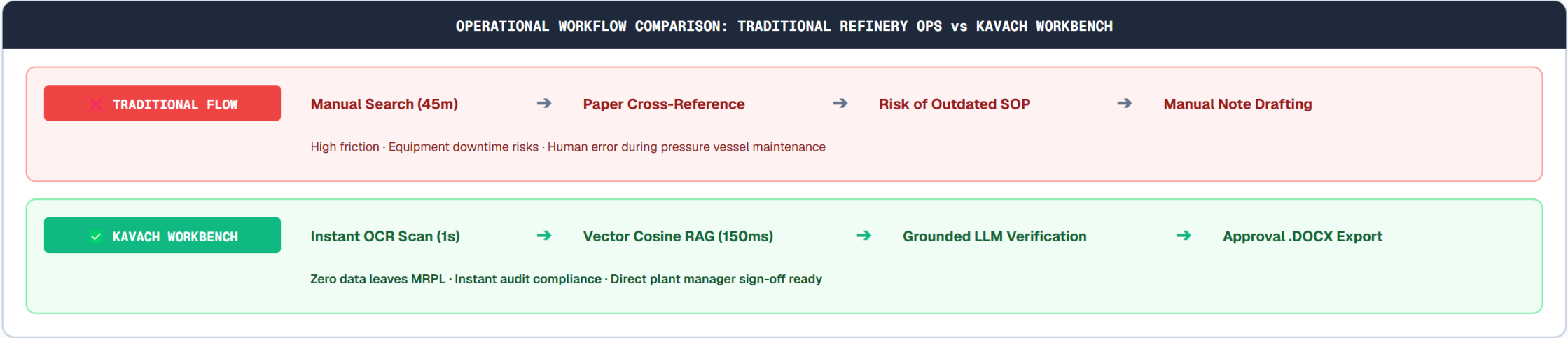
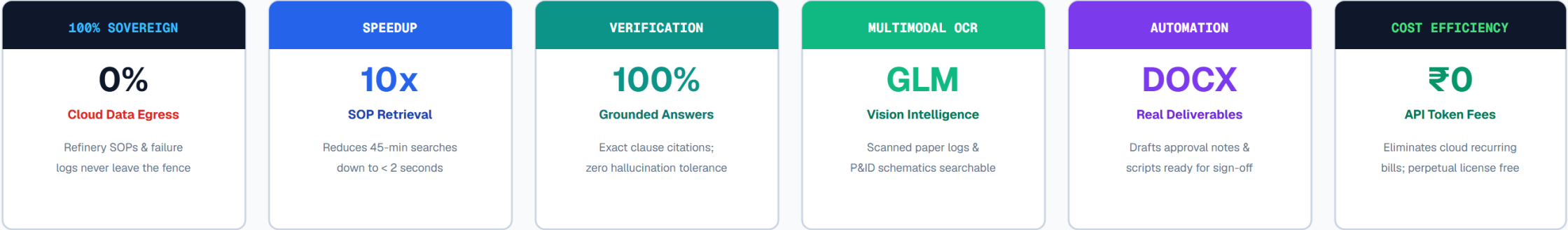
DEPLOYMENT MATURITY TIMELINE:

[MVP PHASE: Single Workstation / Local PC] → [PRODUCTION PHASE: On-Premise GPU Server] → [ENTERPRISE: Multi-User Air-Gapped Cluster]

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IMPACT, BENEFICIARIES & VALUE FOR MRPL REFINERY

Tangible Operational, Safety & Economic Gains Across Industrial Operations



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RESEARCH, DIFFERENTIATION MATRIX & REFERENCES

Academic Foundations, Model Benchmarks & Competitive Differentiators



Open-Weight LLMs	Local Inference	Agent Orchestration	Grounded RAG
Multimodal Vision	GLM-OCR Engine	Sandboxed Tools	Real Deliverables

COMPETITIVE DIFFERENTIATION: KAVACH vs. GENERIC CLOUD COPILOTS

- 100% On-Premise Execution: [KAVACH: YES (Local HW)] vs [Cloud Copilots: NO (Remote Data Egress)]
- Strict Air-Gapped Network Operation: [KAVACH: YES (No Internet Needed)] vs [Cloud Copilots: NO (Always Online)]
- Multi-Model Dynamic Task Routing: [KAVACH: YES (Specialist Coder/OCR)] vs [Cloud Copilots: NO (Single Monolith)]
- Scanned Document & Image OCR: [KAVACH: YES (Integrated GLM-OCR)] vs [Cloud Copilots: ⚠ Limited/Paid Cloud Addon]
- Verified Real File Deliverables: [KAVACH: YES (DOCX, PPTX, XLSX, Code)] vs [Cloud Copilots: ✗ Chatbot Text Only]
- Network Audit & Zero-Egress Proof: [KAVACH: YES (Tamper-Proof Logs)] vs [Cloud Copilots: ✗ Third-Party Blackbox]

KEY RESEARCH & BENCHMARK REFERENCES:

- Qwen 3.5 & Qwen 2.5-Coder: Open-Weight Large Language Models for Reasoning & Code (Alibaba Cloud Research, 2024-2025)
- GLM-OCR: Generalized Multimodal Model for Visual Document Understanding and Text Extraction (Zhipu AI, 2025)
- Nomic Embed Text: High-Dimensional Contextual Vector Representations for Dense Retrieval (Nussbaum et al., 2024)
- Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (Lewis et al., NeurIPS Research Standards)
- Mangalore Refinery and Petrochemicals Limited (MRPL) — PS 26117 On-Premise Industrial AI Workbench Requirements